



A

Mini Project

On

**A MACHINE LEARNING APPROACH FOR TRACKING
AND PREDICTING STUDENT PERFORMANCE
IN DEGREE PROGRAMS**

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

in



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

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A part from the efforts of us, the success of any project depends largely on the encouragement and guidelines of many others. We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project. We take this opportunity to express my profound gratitude and deep regard to my guide.

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ABSTRACT

Accurately predicting students future performance based on their ongoing academic records is crucial for effectively carrying out necessary pedagogical interventions to ensure students' on-time and satisfactory graduation. The proposed method has two major features. First, a bilayered structure comprising multiple base predictors and a cascade of ensemble predictors is developed for making predictions based on students' evolving performance states. Second, a data-driven approach based on latent factor models and probabilistic matrix factorization is proposed to discover course relevance, which is important for constructing

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1. INTRODUCTION

1.1 PROJECT SCOPE

The scope of this project involves developing a machine learning framework to accurately predict student performance in degree programs, enabling educational institutions to proactively support student success and improve retention rates. Unlike traditional models, which often focus narrowly on course-specific outcomes, this project takes a

.1.3 PROJECT FEATURES

The project features a comprehensive machine learning framework designed to predict student performance in degree programs with enhanced accuracy and adaptability. It employs a bilayered structure that integrates multiple base and ensemble predictors, enabling dynamic tracking of student progress over time. By utilizing latent factor models and probabilistic matrix factorization, the system assesses the relevance of individual courses, ensuring that the most informative courses are prioritized in predictions. This data-driven approach analyzes various metrics, including student demographics, academic records, and

LITERATURE SURVEY

Machine learning approaches have become increasingly valuable in the education sector, specifically for tracking and predicting student performance. The ability to monitor student progress and forecast academic outcomes is essential for institutions aiming to enhance student success and retention rates. Historically, tracking relied on manual methods like attendance records and grade books, which provided limited insights into a student's performance trajectory. With the advancement of digital platforms, real-time tracking of engagement, attendance, and assignment completion has become possible. When combined

Datasets are fundamental to developing accurate tracking and prediction models. Publicly available educational datasets from platforms like Coursera and edX are popular for general studies. These datasets provide substantial information about student engagement and course completion but lack the specificity needed for institutional applications. Therefore, many institutions rely on their own data from learning management systems (LMS), grading records, and attendance logs. Institutional datasets are richer in detail, encompassing cumulative grades, participation metrics, and engagement behaviors. Key features typically include academic indicators such as GPA, individual course grades, and

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Implementing machine learning models for tracking and predicting student performance is not without challenges. One of the main issues is data imbalance, where there are fewer students in certain performance categories, such as at-risk or low-achieving groups. Techniques like Synthetic Minority Over-sampling Technique (SMOTE) address this imbalance by creating additional samples in underrepresented categories, ensuring more balanced model predictions. Privacy concerns are also significant, as educational data is sensitive and subject to privacy regulations like GDPR in Europe and FERPA in the United States. These regulations emphasize data anonymization to protect student identities, which

In conclusion, machine learning models have transformed how educational institutions track and predict student performance. Ensemble and hybrid models continue to demonstrate high accuracy and flexibility, while advancements in time-series analysis and natural language processing for analyzing student feedback offer promising future directions. Research continues to explore real-time, privacy-preserving machine learning models, with a focus on including diverse features such as psychosocial data for a holistic view of student progress. As the technology advances, machine learning's potential in education may lead to changes in educational policies, such as data-driven funding,

2. SYSTEM ANALYSIS

SYSTEM ANALYSIS

System Analysis is the important phase in the system development process. The System is studied to the minute details and analyzed. The system analyst plays an important role of an interrogator and dwells deep into the working of the present system. In analysis, a detailed study of these operations performed by the system and their relationships within and outside the system is done. A key question considered here is, “what must be done to solve

2.2 EXISTING SYSTEM

The existing systems for predicting student performance primarily rely on traditional statistical methods and basic machine learning algorithms, often focusing on course-specific outcomes rather than overall degree completion. These approaches typically utilize historical academic data, demographic information, and attendance records to assess student performance. However, they face several limitations: they often do not account for the diverse backgrounds and varying course relevance that significantly influence academic success. Additionally, many existing models are static, failing to incorporate the dynamic

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- **Overreliance on Historical Data:** Existing models often depend heavily on historical data, which may not adequately reflect current trends or changes in educational practices.
- **Limited Predictive Power:** Basic machine learning algorithms used in current systems may not capture the complexity of the factors influencing student performance, resulting in lower predictive accuracy.
- **Complexity of Academic Paths:** Existing systems may struggle to accommodate the complexity of individual academic paths, which can vary widely even among students in the same program.
- **Difficulty in Integrating Real-Time Data:** Many traditional models do not

2.2 PROPOSED SYSTEM

The proposed system introduces an innovative machine learning framework designed to predict student performance in degree programs more accurately and effectively. This approach features a bilayered structure that integrates multiple base predictors with a cascade of ensemble predictors, allowing for dynamic assessments of students' evolving performance states over time. By analyzing various metrics, including historical academic records, demographic data, and behavioral indicators, the system can adapt to changes in student performance and provide timely insights. Additionally, a data-driven methodology

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- **Early Identification of At-Risk Students:** The ability to predict student performance early enables timely interventions, helping to address issues before they become critical.
- **Holistic View of Student Success:** The system analyzes a wide range of metrics, including academic records, demographics, and behavioral data, providing a well-rounded perspective on student performance.
- **Personalized Interventions:** By understanding individual student needs, educators can tailor support strategies to enhance student outcomes effectively.

2.3.1 ECONOMIC FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

2.3.2 TECHNICAL FEASIBILITY

2.4 HARDWARE & SOFTWARE REQUIREMENTS

2.4.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements.

- **Processor:** Intel Core i3

3.ARCHITECTURE

3.1 PROJECT ARCHITECTURE

3.3 USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

3.4 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

3.5 SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

3.6 ACTIVITYDIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

4. IMPLEMENTATION

4.1 SOURCE CODE

View.py:

```
from django.shortcuts import render  
  
from django.template import RequestContext  
  
from django.contrib import messages
```

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```
from sklearn.metrics import recall_score  
  
from sklearn.metrics import f1_score  
  
from sklearn.metrics import accuracy_score  
  
import matplotlib.pyplot as plt  
  
import io  
  
import base64
```

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```
dataset.drop(['FinalResult'], axis = 1,inplace=True)
```

```
X = dataset.values
```

```
sc = StandardScaler()
```

```
X = sc.fit_transform(X)
```

```
indices = np.arange(X.shape[0])
```

```
np.random.shuffle(indices)
```

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```
rf = RandomForestClassifier(n_estimators=1)

rf.fit(X_train, y_train)

predict = rf.predict(X_test)

calculateMetrics("Random Forest", predict, y_test)

svm_cls = svm.SVC(C=1.0, kernel="linear")

svm_cls.fit(X_train, y_train)
```

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```
def PredictPerformance(request):  
  
    if request.method == 'GET':  
  
        return render(request, 'PredictPerformance.html', {})  
  
def PredictPerformanceAction(request):  
  
    if request.method == 'POST':  
  
        global label_encoder, rf, sc, labels, graph_data, gb_cls, xg_cls
```

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```
absent = float(request.POST.get('t17', False).strip())
```

```
score1 = float(request.POST.get('t18', False).strip())
```

```
score2 = float(request.POST.get('t19', False).strip())
```

```
score3 = float(request.POST.get('t20', False).strip())
```

```
data = []
```

```
data.append([gender,age,mother,father,reason,guardian,study,failure,school,family,paid,activity,internet,free,out,health,absent,score1,score2,score3])
```

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```
predict = labels[predict]

graph_data.append(predict)

status = ""

if predict == "Poor":

    status = "Warning! Need more focus & hardwork"

output = "Roll No : "+roll_no+"<br/>Overall Predicted Performance ==>
```

```
def TrainML(request):  
  
    if request.method == 'GET':  
  
        output = "  
  
        output+=<table border=1 align=center width=100%><tr><th><font size=""  
color="black">Algorithm Name</th><th><font size=""  
color="black">Accuracy</th><th><font size="" color="black">Precision</th>  
  
        output+=<th><font size="" color="black">Recall</th><th><font size=""
```

```
['XGBoost','Precision',precision[3]],['XGBoost','Recall',recall[3]],['XGBoost','F1
Score',fscore[3]],['XGBoost','Accuracy',accuracy[3]],

],columns=['Algorithms','Metrics','Value'])

df.pivot_table(index="Algorithms", columns="Metrics",
values="Value").plot(kind='bar', figsize=(8, 4))

plt.title("All Algorithms Performance Graph")

plt.tight_layout()
```

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```
return render(request, 'LoadDataset.html', {})
```

```
def AdminLoginAction(request):
```

```
    if request.method == 'POST':
```

```
        global uname
```

```
        username = request.POST.get('username', False)
```

```
        password = request.POST.get('password', False)
```

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```
dataset = dataset.values
```

```
output='<table border=1 align=center width=100%><tr>'
```

```
for i in range(len(columns)):
```

```
    output += '<th><font size="" color="black">'+columns[i]+'</th>'
```

```
output += '</tr>'
```

```
for i in range(len(dataset)):
```

RESULTS:

6. TESTING

6.1 INTRODUCTION TO TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover conceivable fault or weakness in a work product. It provides a way to check the functionality of components, subassemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are

6.2.3 FUNCTIONAL TESTING

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.

TABLE NO- 6.3.2 CLASSIFICATION

Test case ID	Testcase name	Purpose	Input	Output
1	Classification test 1	To check if the classifier performs its task	A Student's details are given	Student's performance is predicted as good.

7. PROJECT CONCLUSION & FUTURE SCOPE

7.1 PROJECT CONCLUSION

In conclusion, the proposed machine learning framework for predicting student performance in degree programs represents a significant advancement in educational data analysis. By effectively addressing key challenges such as diverse student backgrounds, course relevance, and the dynamic nature of academic progress, the system enhances predictive accuracy and adaptability. Its bilayered structure and innovative methods for

8. BIBLIOGRAPHY

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These references provide a foundation for understanding the current landscape of research in predicting student performance and the application of machine learning techniques in education.